

Deep Generative Models

Lecture 2

Roman Isachenko

Moscow Institute of Physics and Technology
Yandex School of Data Analysis

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Recap of Previous Lecture

We're given a **finite** number of i.i.d. samples $\{\mathbf{x}_i\}_{i=1}^n \subset \mathbb{R}^m$ drawn from an **unknown** distribution $p_{\text{data}}(\mathbf{x})$.

Objective

Our aim is to estimate a distribution $p_{\text{data}}(\mathbf{x})$ that allows us to:

- ▶ Generate new samples from $p_{\text{data}}(\mathbf{x})$ (sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$) — **generation**;
- ▶ Evaluate $p_{\text{data}}(\mathbf{x})$ on novel data (answering “How likely is an object $\mathbf{x}$?”) — **density estimation**.

What is a Divergence?

- ▶ $D(\pi \| p) \geq 0$ for all $\pi, p \in \mathcal{P}$
- ▶ $D(\pi \| p) = 0$ if and only if $\pi \equiv p$

Divergence Minimization Problem

$$\min_{\theta} D(p_{\text{data}} \| p_{\theta})$$

Recap of Previous Lecture

Forward KL

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \int p_{\text{data}}(\mathbf{x}) \log \frac{p_{\text{data}}(\mathbf{x})}{p_{\theta}(\mathbf{x})} d\mathbf{x} \rightarrow \min_{\theta}$$

Reverse KL

$$D_{\text{KL}}(p_{\theta} \| p_{\text{data}}) = \int p_{\theta}(\mathbf{x}) \log \frac{p_{\theta}(\mathbf{x})}{p_{\text{data}}(\mathbf{x})} d\mathbf{x} \rightarrow \min_{\theta}$$

Maximum Likelihood Estimation (MLE)

Let $\{\mathbf{x}_i\}_{i=1}^n$ be i.i.d. observed samples.

$$\theta^* = \arg \max_{\theta} \prod_{i=1}^n p_{\theta}(\mathbf{x}_i) = \arg \max_{\theta} \sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i).$$

Maximum likelihood estimation is equivalent to minimizing a Monte Carlo estimate of the forward KL divergence.

Recap of Previous Lecture

Likelihood as a Product of Conditionals

For $\mathbf{x} = (x_1, \dots, x_m)$, $\mathbf{x}_{1:j} = (x_1, \dots, x_j)$,

$$p_{\theta}(\mathbf{x}) = \prod_{j=1}^m p_{\theta}(x_j | \mathbf{x}_{1:j-1}); \quad \log p_{\theta}(\mathbf{x}) = \sum_{j=1}^m \log p_{\theta}(x_j | \mathbf{x}_{1:j-1})$$

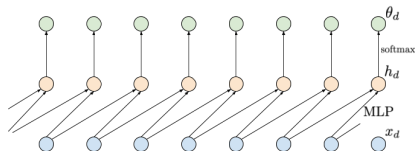
$$\theta^* = \arg \max_{\theta} \sum_{i=1}^n \left[\sum_{j=1}^m \log p_{\theta}(x_{ij} | \mathbf{x}_{i,1:j-1}) \right]$$

Sampling (Ancestral)

1. Sample $\hat{x}_1 \sim p_{\theta}(x_1)$, $\hat{x}_2 \sim p_{\theta}(x_2 | \hat{x}_1)$, $\dots$,
 $\hat{x}_m \sim p_{\theta}(x_m | \hat{\mathbf{x}}_{1:m-1})$.
2. Return $\hat{\mathbf{x}} = (\hat{x}_1, \hat{x}_2, \dots, \hat{x}_m)$.

Recap of Previous Lecture

Autoregressive Models: MLP



Autoregressive Models: ImageGPT

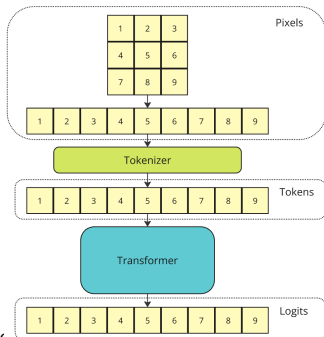
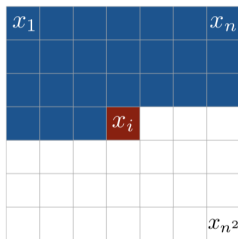


Image credit: <https://jmtomczak.github.io/blog/>,
Oord A., Kalchbrenner N., Kavukcuoglu K. Pixel Recurrent Neural Networks, 2016
Chen M. et al. Generative Pretraining from Pixels, 2020

Outline

1. Normalizing Flows (NF)

2. NF Examples

- Linear NF

- Gaussian Autoregressive NF

- Coupling Layer (RealNVP)

3. Latent Variable Models (LVM)

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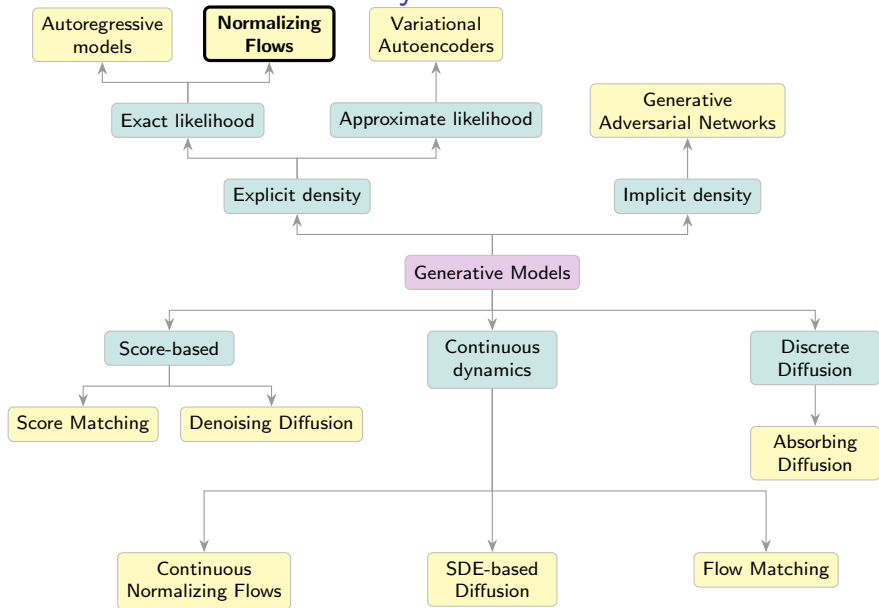
Linear NF

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Coupling Layer (RealNVP)

3. Latent Variable Models (LVM)

Generative Models Taxonomy



Normalizing Flows: Prerequisites

Jacobian Matrix

Let $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a differentiable function.

$$\mathbf{z} = \mathbf{f}(\mathbf{x}), \quad \mathbf{J} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial z_1}{\partial x_1} & \cdots & \frac{\partial z_1}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_m}{\partial x_1} & \cdots & \frac{\partial z_m}{\partial x_m} \end{pmatrix} \in \mathbb{R}^{m \times m}$$

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Change of Variables Theorem (CoV)

Let $\mathbf{x} \in \mathbb{R}^m$ be a random vector with density $p(\mathbf{x})$, and let $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a C^1 -diffeomorphism ($\mathbf{f}$ and $\mathbf{f}^{-1}$ are continuously differentiable mappings). If $\mathbf{z} = \mathbf{f}(\mathbf{x})$, then

$$p(\mathbf{x}) = p(\mathbf{z}) |\det(\mathbf{J}_{\mathbf{f}})| = p(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right|$$

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Jacobian Determinant

Inverse Function Theorem

If the function $\mathbf{f}$ is invertible and its Jacobian is continuous and non-singular, then

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- ▶ $\mathbf{f}_{\theta}(\mathbf{x})$ is a parameterized transformation.

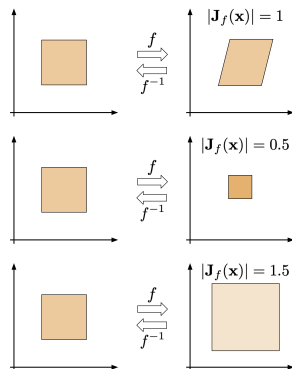
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- ▶ $\mathbf{f}_{\theta}(\mathbf{x})$ is a parameterized transformation.
- ▶ The determinant of the Jacobian $\mathbf{J} = \frac{\partial \mathbf{f}_{\theta}(\mathbf{x})}{\partial \mathbf{x}}$ quantifies how the volume is changed by the transformation.



Fitting Normalizing Flows

MLE Problem

$$p_{\theta}(\mathbf{x}) = p(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right| = p(\mathbf{f}_{\theta}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}_{\theta}(\mathbf{x})}{\partial \mathbf{x}} \right) \right|$$

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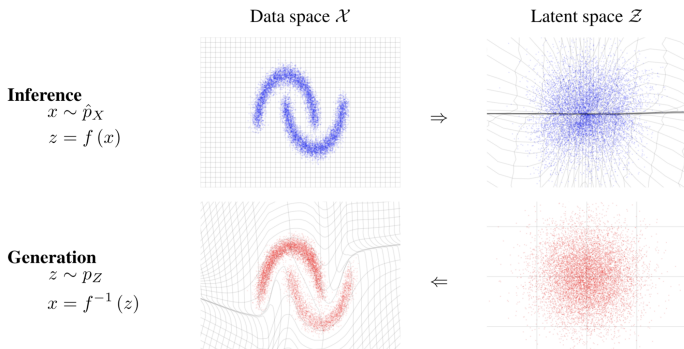
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Composition of Normalizing Flows

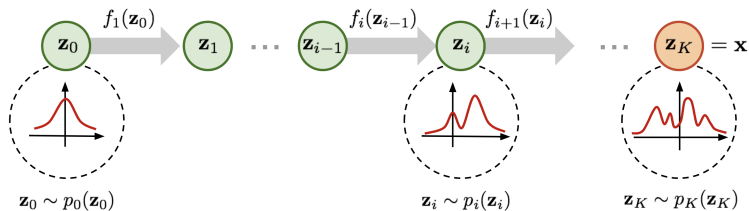
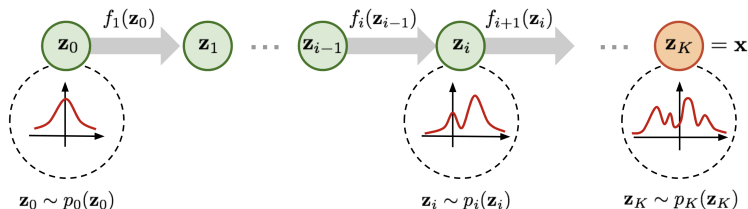


Image credit: <https://lilianweng.github.io/lil-log/2018/10/13/flow-based-deep-generative-models.html>

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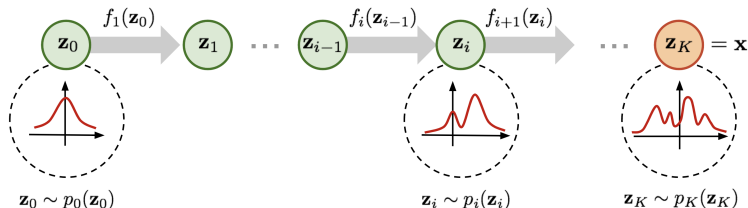


Theorem

If every $\{\mathbf{f}_k\}_{k=1}^K$ satisfies the conditions of the change-of-variables theorem, then the composition $\mathbf{f}(\mathbf{x}) = \mathbf{f}_K \circ \dots \circ \mathbf{f}_1(\mathbf{x})$ also satisfies them.

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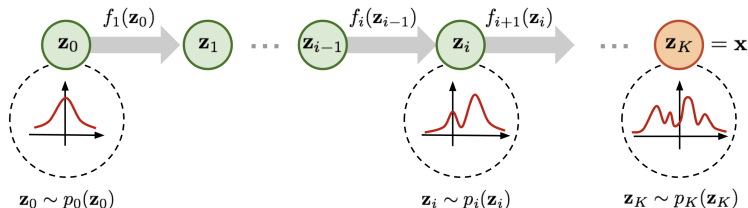


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Normalizing Flows (NF)

$$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{f}_{\theta}(\mathbf{x})) + \log |\det(\mathbf{J}_{\mathbf{f}})|$$

Definition

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- ▶ **Normalizing** refers to mapping samples from $p_{\text{data}}(\mathbf{x})$ to a base distribution $p(\mathbf{z})$.
- ▶ **Flow** describes the sequence of transformations that maps samples from $p(\mathbf{z})$ to the target, more complex distribution.

$$\mathbf{z} = \mathbf{f}_K \circ \dots \circ \mathbf{f}_1(\mathbf{x}); \quad \mathbf{x} = \mathbf{f}_1^{-1} \circ \dots \circ \mathbf{f}_K^{-1}(\mathbf{z})$$

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Log-Likelihood

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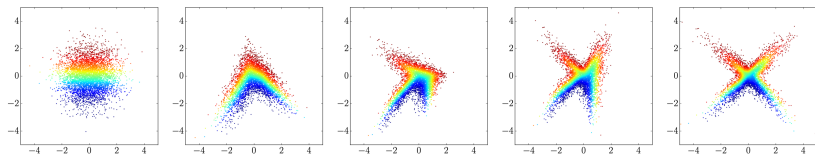
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Note: Here we consider only continuous random variables.
Gutierrez, 2019

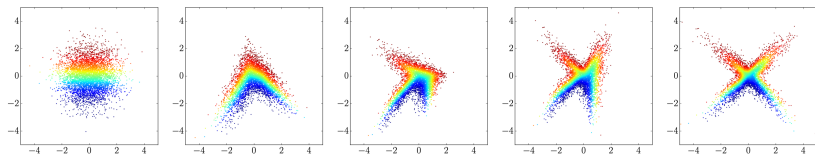
Normalizing Flows

Example: 4-Step NF



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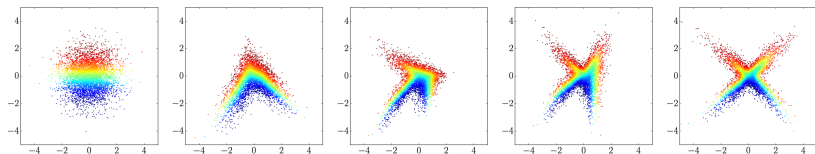
NF Log-Likelihood

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What's the computational complexity of evaluating this determinant?

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Requirements

- ▶ Efficient computation of the Jacobian $\mathbf{J}_{\mathbf{f}} = \frac{\partial \mathbf{f}_{\theta}(\mathbf{x})}{\partial \mathbf{x}}$
- ▶ Efficient inversion of the transformation $\mathbf{f}_{\theta}(\mathbf{x})$

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The principal computational challenge is evaluating the Jacobian determinant.

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What is $\det(\mathbf{J})$ in These Cases?

Consider a linear layer $\mathbf{z} = \mathbf{W}\mathbf{x}$, $\mathbf{W} \in \mathbb{R}^{m \times m}$.

1. $\mathbf{z}$ is a permutation of $\mathbf{x}$.

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3. z_j depends only on $\mathbf{x}_{1:j}$ (autoregressive dependency).

Linear Normalizing Flows

$$\mathbf{z} = \mathbf{f}_{\theta}(\mathbf{x}) = \mathbf{W}\mathbf{x}, \quad \mathbf{W} \in \mathbb{R}^{m \times m}, \quad \theta = \mathbf{W}, \quad \mathbf{J}_{\mathbf{f}} = \mathbf{W}$$

In general, matrix inversion has computational complexity $O(m^3)$.

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Invertibility

- ▶ Diagonal matrix: $O(m)$.
- ▶ Triangular matrix: $O(m^2)$.
- ▶ Directly parameterizing all invertible matrices in a continuous way is infeasible
(there is no surjective function from $\mathbb{R}^{m^2}$ to the set of all invertible matrices of size $m \times m$).

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Matrix Decompositions

► LU Decomposition:

$$\mathbf{W} = \mathbf{P}\mathbf{L}\mathbf{U},$$

where $\mathbf{P}$ is a permutation matrix, $\mathbf{L}$ is lower triangular with positive diagonal, and $\mathbf{U}$ is upper triangular with positive diagonal.

Linear Normalizing Flows

$$\mathbf{z} = \mathbf{f}_{\theta}(\mathbf{x}) = \mathbf{W}\mathbf{x}, \quad \mathbf{W} \in \mathbb{R}^{m \times m}, \quad \theta = \mathbf{W}, \quad \mathbf{J}_{\mathbf{f}} = \mathbf{W}$$

Matrix Decompositions

► LU Decomposition:

$$\mathbf{W} = \mathbf{P}\mathbf{L}\mathbf{U},$$

where $\mathbf{P}$ is a permutation matrix, $\mathbf{L}$ is lower triangular with positive diagonal, and $\mathbf{U}$ is upper triangular with positive diagonal.

► QR Decomposition:

$$\mathbf{W} = \mathbf{Q}\mathbf{R},$$

where $\mathbf{Q}$ is orthogonal, and $\mathbf{R}$ is upper triangular with positive diagonal.

Linear Normalizing Flows

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► QR Decomposition:

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Decomposition is performed only at initialization; the decomposed matrices ($\mathbf{P}$, $\mathbf{L}$, $\mathbf{U}$ or $\mathbf{Q}$, $\mathbf{R}$) are optimized during training.

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Gaussian Autoregressive Model

Consider the autoregressive model:

$$p_{\theta}(\mathbf{x}) = \prod_{j=1}^m p_{\theta}(x_j | \mathbf{x}_{1:j-1}), \quad p_{\theta}(x_j | \mathbf{x}_{1:j-1}) = \mathcal{N}(\mu_{j,\theta}(\mathbf{x}_{1:j-1}), \sigma_{j,\theta}^2(\mathbf{x}_{1:j-1}))$$

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Generative Direction

$$x_j = \sigma_{j,\theta}(\mathbf{x}_{1:j-1}) \cdot z_j + \mu_{j,\theta}(\mathbf{x}_{1:j-1}), \quad z_j \sim \mathcal{N}(0, 1)$$

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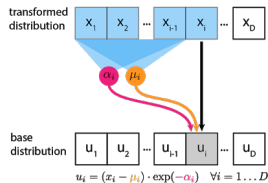
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- ▶ The Jacobian matrix of this transformation is triangular.

Gaussian Autoregressive NF

Forward Transformation: $\mathbf{f}_\theta(\mathbf{x})$

$$\mathbf{z} = \mathbf{f}_\theta(\mathbf{x})$$

$$z_j = \frac{x_j - \mu_{j,\theta}(\mathbf{x}_{1:j-1})}{\sigma_{j,\theta}(\mathbf{x}_{1:j-1})}$$



Gaussian Autoregressive NF

Forward Transformation: $\mathbf{f}_\theta(\mathbf{x})$

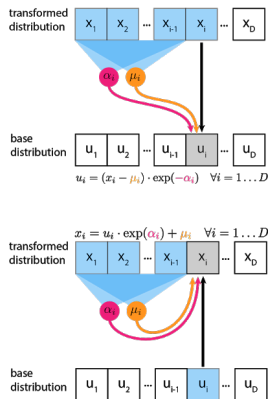
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Gaussian Autoregressive NF

Forward Transformation: $\mathbf{f}_\theta(\mathbf{x})$

$$\mathbf{z} = \mathbf{f}_\theta(\mathbf{x})$$

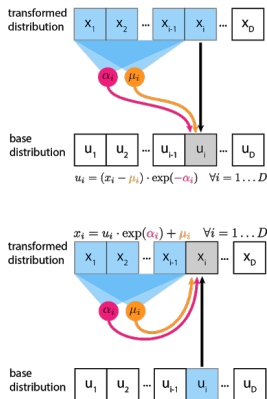
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- ▶ Sampling must be done sequentially, but density estimation can be parallelized.
- ▶ The forward KL divergence is a natural objective for training.



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RealNVP

Split $\mathbf{x}$ and $\mathbf{z}$ into two parts:

$$\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2] = [\mathbf{x}_{1:d}, \mathbf{x}_{d+1:m}]; \quad \mathbf{z} = [\mathbf{z}_1, \mathbf{z}_2] = [\mathbf{z}_{1:d}, \mathbf{z}_{d+1:m}]$$

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Coupling Layer

$$\begin{cases} \mathbf{x}_1 = \mathbf{z}_1 \\ \mathbf{x}_2 = \mathbf{z}_2 \odot \sigma_{\theta}(\mathbf{z}_1) + \mu_{\theta}(\mathbf{z}_1) \end{cases}$$

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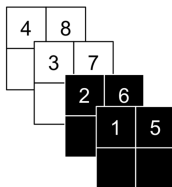
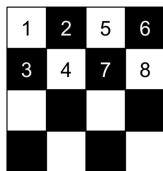
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Image Partitioning



- ▶ Checkerboard ordering corresponds to masking.
- ▶ Channelwise ordering relies on splitting.

RealNVP

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In both training and sampling, only a single forward pass is needed!

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Jacobian

$$\det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) = \det \begin{pmatrix} \mathbf{I}_d & 0_{d \times m-d} \\ \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_1} & \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_2} \end{pmatrix}$$

RealNVP

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Gaussian AR NF

$$\begin{aligned} \mathbf{x} = \mathbf{f}_{\theta}^{-1}(\mathbf{z}) &\Rightarrow \mathbf{x}_j = \sigma_{j,\theta}(\mathbf{x}_{1:j-1}) \cdot \mathbf{z}_j + \mu_{j,\theta}(\mathbf{x}_{1:j-1}) \\ \mathbf{z} = \mathbf{f}_{\theta}(\mathbf{x}) &\Rightarrow \mathbf{z}_j = (\mathbf{x}_j - \mu_{j,\theta}(\mathbf{x}_{1:j-1})) \cdot \frac{1}{\sigma_{j,\theta}(\mathbf{x}_{1:j-1})}. \end{aligned}$$

How can the RealNVP layer be derived as a special instance of the Gaussian autoregressive NF?

Dinh L., Sohl-Dickstein J., Bengio S. Density Estimation Using Real NVP, 2016

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Bayesian Framework

Bayes' Theorem

$$p(\theta|\mathbf{x}) = \frac{p(\mathbf{x}|\theta)p(\theta)}{p(\mathbf{x})} = \frac{p(\mathbf{x}|\theta)p(\theta)}{\int p(\mathbf{x}|\theta)p(\theta)d\theta}$$

- ▶ $\mathbf{x}$: observed variables;
- ▶ θ : unknown latent variables/parameters;
- ▶ $p_{\theta}(\mathbf{x}) = p(\mathbf{x}|\theta)$: likelihood;
- ▶ $p(\mathbf{x}) = \int p(\mathbf{x}|\theta)p(\theta)d\theta$: evidence;
- ▶ $p(\theta)$: prior distribution;
- ▶ $p(\theta|\mathbf{x})$: posterior distribution.

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Interpretation

- ▶ We begin with unknown variables θ and a prior belief $p(\theta)$.
- ▶ Once data $\mathbf{x}$ is observed, the posterior $p(\theta|\mathbf{x})$ incorporates both prior beliefs and evidence from the data.

Bayesian Framework

Consider the case where the unobserved variables θ are model parameters (i.e., θ are random variables).

- ▶ $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^n$: observed samples;
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$$p(\theta|\mathbf{X}) = \frac{p(\mathbf{X}|\theta)p(\theta)}{p(\mathbf{X})} = \frac{p(\mathbf{X}|\theta)p(\theta)}{\int p(\mathbf{X}|\theta)p(\theta)d\theta}$$

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If the evidence $p(\mathbf{X})$ is intractable (due to high-dimensional integration), the posterior cannot be computed exactly.

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Maximum a Posteriori (MAP) Estimation

$$\theta^* = \arg \max_{\theta} p(\theta|\mathbf{X}) = \arg \max_{\theta} (\log p(\mathbf{X}|\theta) + \log p(\theta))$$

Latent Variable Models (LVM)

Maximum Likelihood Estimation (MLE) Problem

$$\theta^* = \arg \max_{\theta} p_{\theta}(\mathbf{X}) = \arg \max_{\theta} \prod_{i=1}^n p_{\theta}(\mathbf{x}_i) = \arg \max_{\theta} \sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i).$$

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Introduce a latent variable $\mathbf{z}$ for each observed sample $\mathbf{x}$:

$$p_{\theta}(\mathbf{x}, \mathbf{z}) = p_{\theta}(\mathbf{x}|\mathbf{z})p(\mathbf{z}); \quad \log p_{\theta}(\mathbf{x}, \mathbf{z}) = \log p_{\theta}(\mathbf{x}|\mathbf{z}) + \log p(\mathbf{z}).$$

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- ▶ $p(\mathbf{z})$ is the prior distribution over the latent variable.
- ▶ $p_{\theta}(\mathbf{x}|\mathbf{z})$ is the **decoder** distribution.
- ▶ Both $p_{\theta}(\mathbf{x}|\mathbf{z})$ and $p(\mathbf{z})$ are usually much simpler than $p_{\theta}(\mathbf{x})$.

Latent Variable Models (LVM)

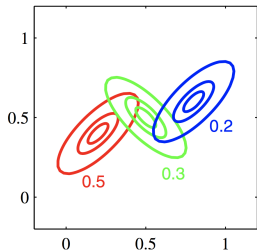
$$\log p_{\theta}(\mathbf{x}) = \log \int p_{\theta}(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z} \rightarrow \max_{\theta}$$

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Examples

Mixture of Gaussians



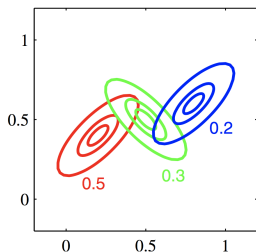
- ▶ $p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_{\mathbf{z}}, \boldsymbol{\Sigma}_{\mathbf{z}})$
- ▶ $p(\mathbf{z}) = \text{Cat}(\boldsymbol{\pi})$

Latent Variable Models (LVM)

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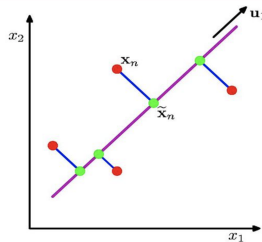
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PCA Model



- ▶ $p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I})$
- ▶ $p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$

Summary

- ▶ The CoV theorem provides a method for computing a random variable's density under an invertible transformation.
- ▶ Normalizing flows transform a simple base distribution into a complex one via a sequence of invertible mappings, each with efficient Jacobian determinants.
- ▶ Linear NFs capture invertible matrices by using matrix decompositions.
- ▶ Gaussian autoregressive NFs are AR models with triangular Jacobians.
- ▶ The RealNVP coupling layer provides an efficient normalizing flow (a special case of AR NF), supporting fast inference and sampling.
- ▶ LVMs introduce latent representations for observed data, building a complex $p_{\theta}(\mathbf{x})$ from much simpler $p_{\theta}(\mathbf{x}|\mathbf{z})$ and $p(\mathbf{z})$.